

Nicolás Cerrato Antón

SOFTWARE ENGINEER · AGENTIC SYSTEMS · MARKETS

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Software Engineer with a strong foundation in Computer Science and an international academic background (Spain / Germany). Focused on **agentic systems**, **full-stack web development**, **algorithmic trading**, and **blockchain protocols**. Builds production-grade tooling across TypeScript, Python, and Solidity — from LLM-powered trading cockpits and agent orchestrators to editorial AI and on-chain monetary systems. Trilingual professional (English, German, Spanish).

01 TECHNICAL PROJECTS

Agentic Systems

TypeScript · Bun · WebSocket · SQLite

- Designed multi-agent architectures with DAG-based scheduling, dependency resolution, cancellation propagation, and event-sourced audit trails.
- Built deterministic execution layers that constrain LLM-generated intent — separating what the agent decides from what the system allows.
- Implemented real-time dashboards and terminal UIs for monitoring agent state, live output, and historical replay.

AI Integration & LLM Tooling

OpenAI · Anthropic · FastAPI · Python

- Integrated streaming LLM responses with structured output parsing across OpenAI and Anthropic models.
- Built embedding pipelines for semantic search and automatic relationship inference over user-generated content.
- Developed editorial and rewriting workflows with inline diff rendering and interactive acceptance flows.

Full-Stack Web Applications

React · Hono · Next.js · Tailwind

- Built production-grade SPAs and full-stack applications with real-time data, complex state management, and rich interactive UIs.
- Designed and implemented REST and WebSocket APIs with typed contracts shared across monorepo packages.

Blockchain & Protocol Design

Solidity · Smart Contracts

- Designed on-chain monetary mechanisms with supply-sensitive stability and tranche-based risk exposure.
- Approached protocol design as a systems engineering problem: incentive alignment, edge-case resilience, and deterministic behavior under adversarial conditions.

02 EDUCATION

B.Sc. Computer Engineering

Universidad de Málaga 2022 – 2024

B.Sc. Computer Engineering (First Cycle)

Albert-Ludwigs-Universität Freiburg 2020 – 2022

Coursework: Machine Learning, Smart Contracts, System Programming.

Abitur — Bilingual Diploma

Deutsche Schule Las Palmas 2006 – 2018

03 LANGUAGES

Spanish Native

German Native / Bilingual

English C2 · Cambridge Certificate

04 TECHNICAL SKILLS

LANGUAGES

TypeScript Python Go C JavaScript Solidity

BACKEND

Hono FastAPI Node.js Express SQLite

WebSocket

FRONTEND

React Next.js Tailwind React Flow MapLibre GL

AI / AGENTS

OpenAI API Anthropic API LLM Agents ReAct

Embeddings

BLOCKCHAIN

Solidity Smart Contracts Algo Stablecoins

TOOLS

Bun Git Docker Linux Bash Vercel